GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form A · 24 points

Name_____
Date_____
Period_____

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 10 cm in its scale copy. Find the scale factor. *1 pt*

2. A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

3. A figure is scaled by a scale factor of 3. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. Each side is 3 times as long.
- ☐ B. Each angle is 3 times as large.
- ☐ C. The perimeter is 3 times as long.
- ☐ D. The area is 3 times as large.
- ☐ E. The area is 9 times as large.

4. A 3 cm by 4 cm rectangle is scaled by k = 3.

a. Find the copy's perimeter. *1 pt*

b. Find the copy's area. *1 pt*

5. A figure has area 8 cm^2. Its scale copy has area 72 cm^2. What is the scale factor? *1 pt*

6. A floor plan is drawn at ¼ in : 1 ft. A room is 3 in by 3.5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

b. Find the room's real area. *1 pt*

7. A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

8. A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 9 in? *1 pt*

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 70 cm. *1 pt*

- ☐ A. 70 × 0.39
- ☐ B. 70 × 2.54
- ☐ C. 70 ÷ 2.54
- ☐ D. 70 ÷ 0.39

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 4 yd? *1 pt*

12. 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

13. 1 gal = 3.785 L. How many liters is 5 gal? *1 pt*

14. 1 ft = 0.305 m. A patio is 20 ft^2 in area. Which is closest to its area in square meters? *1 pt*

- ○ A. 6.1 m^2
- ○ B. 0.093 m^2
- ○ C. 1.86 m^2
- ○ D. 12.2 m^2

15. Rate on Nov 5: 1 US dollar = 0.92 euros.

a. How many euros is $60? *1 pt*

b. How many US dollars is 23 euros? *1 pt*

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 26 miles per gallon and its tank holds 15 gal.

a. How far does it go on a full tank? *1 pt*

b. It makes a 1,000-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

17. A recipe for 4 servings uses $1\frac{1}{2}$ cups of rice.

a. How much rice for 10 servings? *1 pt*

b. With 6 cups on hand, how many servings can be made? *1 pt*

18. Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

19. 1 mi = 1.61 km. A drive is 200 km long. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*